

HRSG 3D AUTO MODELING FROM 2D DWG

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Dedicated pipeline that builds HRSG stack / casing / platform / PPA steel (100–403) into E3D and places nozzles / equipment.

Key Purpose

Why this program

1. Turn 2D steel DXF into a 3D E3D model

– Auto-classifies each drawing (stack / casing) and runs the matching 100-403 generators

→ **Effect:** Replaces hours of manual re-modelling with a one-click build and removes transcription error

2. One DXF pick builds the whole structure

– Beam, handrail, grating, main shell and radial steel merged into a single ready-to-run macro

→ **Effect:** Delivers a consistent, complete model in one run instead of piecing parts together by hand

3. Place nozzles and equipment automatically

– Reads shell / door / damper geometry and writes E3D EQUIPMENT + NOZZLE elements at the origin grid

→ **Effect:** Skips tedious manual primitive placement and keeps every part on the correct column-grid origin

4. Portable, idempotent macro output

– Single reusable ZONE with an origin marker; re-running moves rather than duplicates elements

Required inputs : One or more Steel Structure 2D DXF files

Main Window – Item Guide

The screenshot shows the 'HRSR Modeling Option' dialog box. It has a title bar with a gear icon and the text 'HRSR Modeling Option'. The main area is divided into several sections. At the top, there's a section for 'Select Steel Structure 2D dxf File' with a large empty box (2) and buttons 'Select' (3), 'Remove Selected', 'Clear All', and 'Open'. Below this is the 'Aid Steel Size Expression' section with buttons 'Symbol', 'Real Size', and 'Off' (4). To the right is the 'Hand Rail Level' section with buttons '2 Level' (5) and '3 Level'. The 'Aid Expression' section has three checked checkboxes: 'Col No', 'Red Center', and 'Section' (6). The 'HRSR Modeling' section has four tabs: 'Stack' (7), 'Casing & Inlet Duct', 'Platform', and 'PPA'. Below the tabs are input fields for 'Hand Rail Height' (7, value 1232), 'Macro File Create' (8, value '(DXF 도면과 동일 위치)'), 'Stru Ori. North/East' (9, values 0 and 0), and 'Location Plan Column' (10, value 'North(V):East'). At the bottom are 'Apply' (12) and 'Cancel' buttons (11).

1. General tab
2. Select Steel Structure 2D DXF file list
3. Select / Remove Selected / Clear All / Open
4. Aid Steel Size Expression: Symbol / Real Size / Off
5. Hand Rail Level: 2 Level / 3 Level
6. Aid Expression: Col No / Red Center / Section
7. HRSR Modeling: Stack / Casing & Inlet Duct / Platform / PPA
8. Hand Rail Height
9. Macro File Create (output folder)
10. Stru Ori. North / East origin coordinates
11. Location Plan Column mapping
12. Apply / Cancel

Processing Steps

What 'Apply' runs, in order

1. Read each DXF and classify it (stack / casing / unclassified)
2. Place origin marker at the column-grid N0 E0 origin
3. Generate beam / platform steel sections
4. Generate handrail primitives
5. Generate grating panels
6. Generate main shell or casing frame columns
7. Generate radial / curved platform steel
8. Place nozzles and equipment from shell / door / damper geometry
9. Merge every part into one E3D macro under a single ZONE

Overview

What it does

1. Dedicated pipeline that builds HRSG stack
2. Casing
3. Platform
4. PPA steel (100–403) into E3D and places nozzles
5. Equipment

Category: HRSG 3D Auto Modeling from 2D Dwg

Folder: HRSG

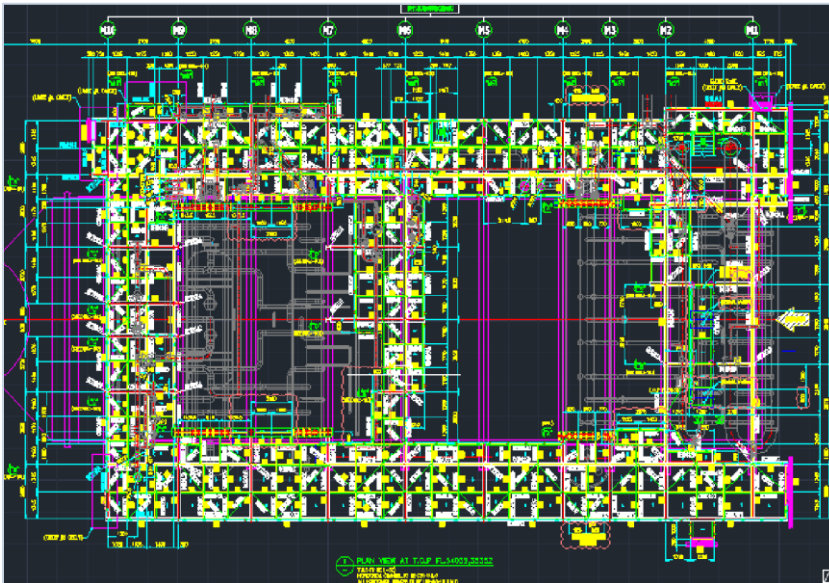
Main scripts: Ace_hrsg_runall.PY

Scripts / Status: 22 scripts · Operational

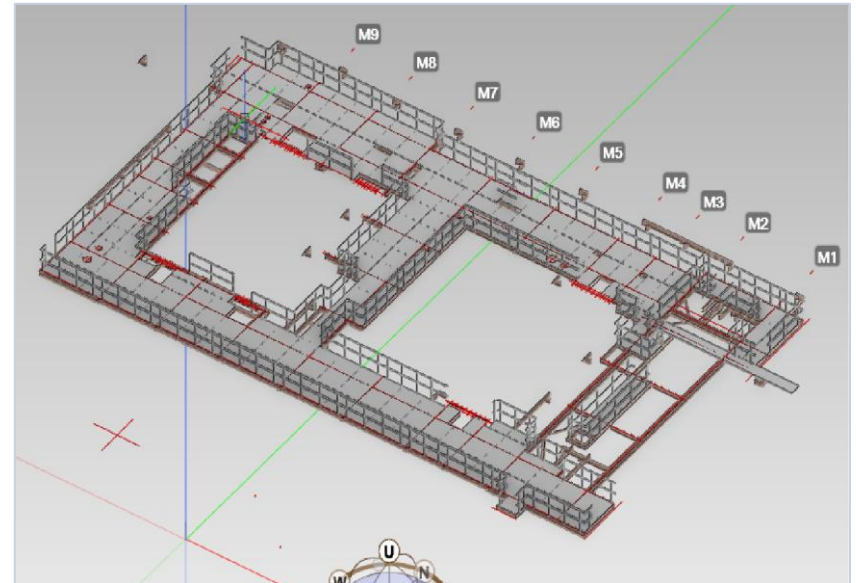
Tags: E3D · HRSG · Stack · Casing · Nozzle

Output Samples

Output Samples (1/7)



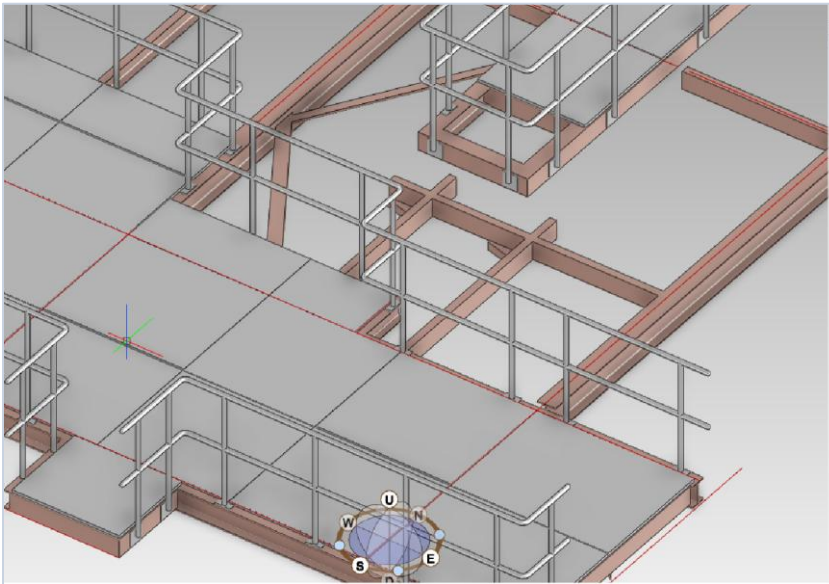
Platform 2D Input



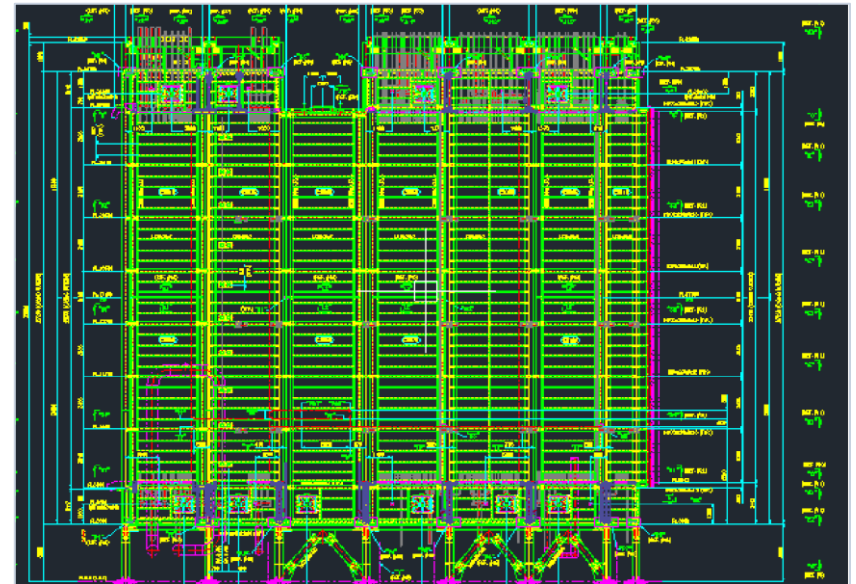
Platform 3D Auto Output

Output Samples

Output Samples (2/7)



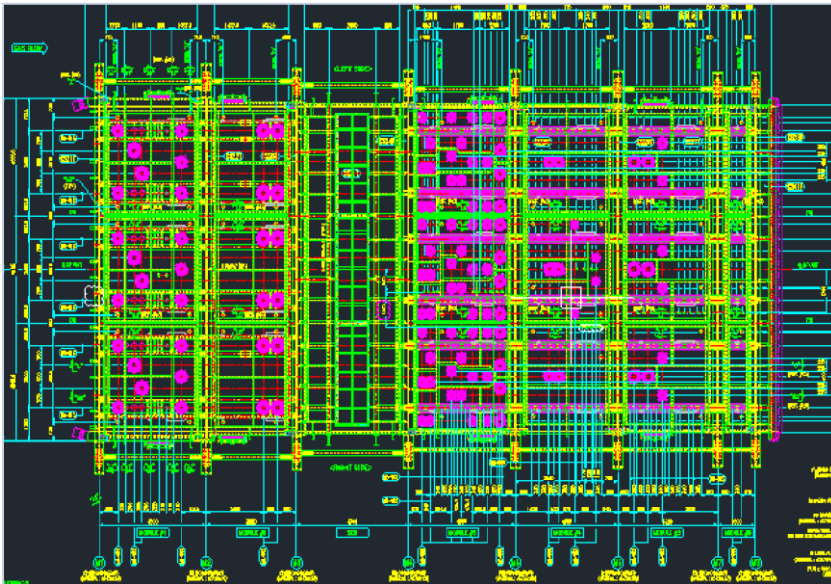
Platform 3D Auto Output (detail)



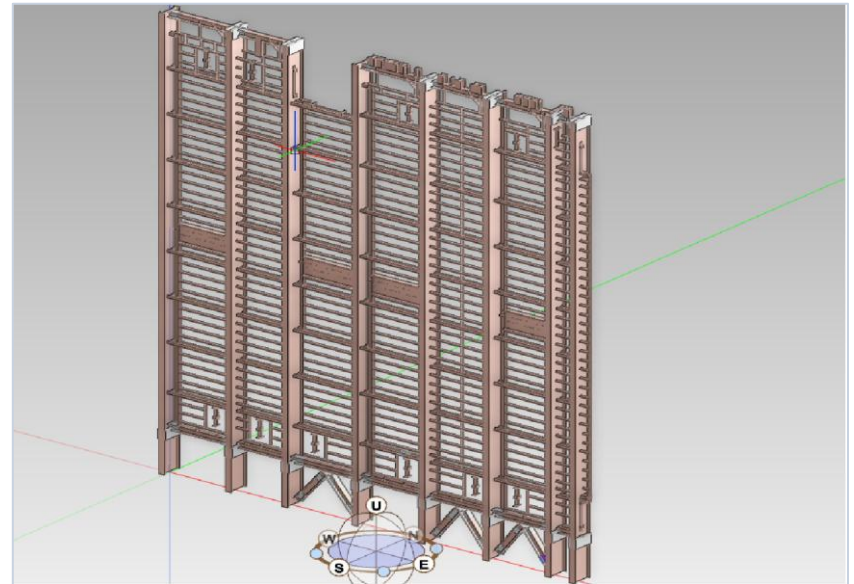
CASING INPUT1

Output Samples

Output Samples (3/7)



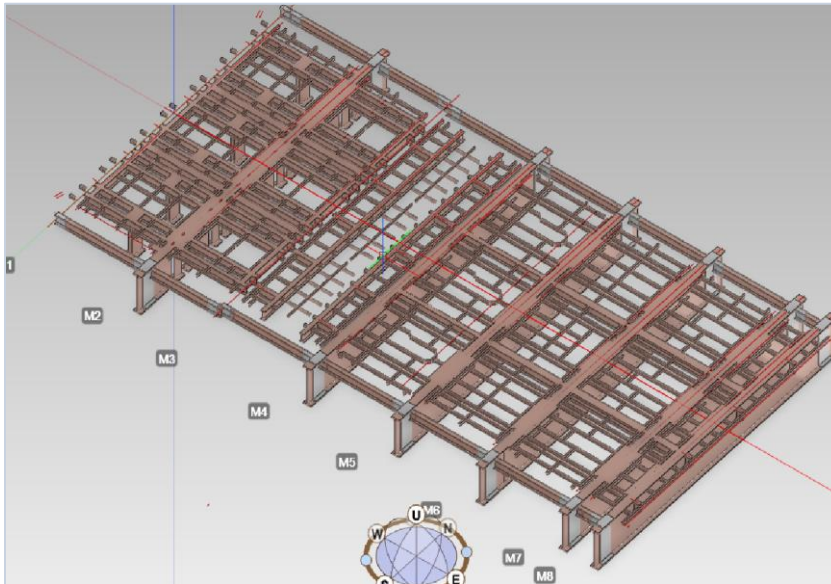
CASING INPUT2



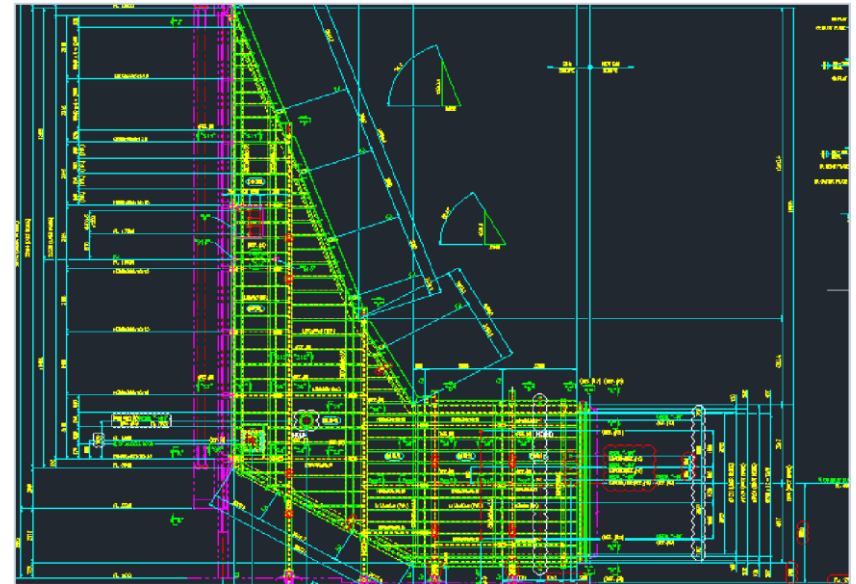
CASING OUTPUT1

Output Samples

Output Samples (4/7)



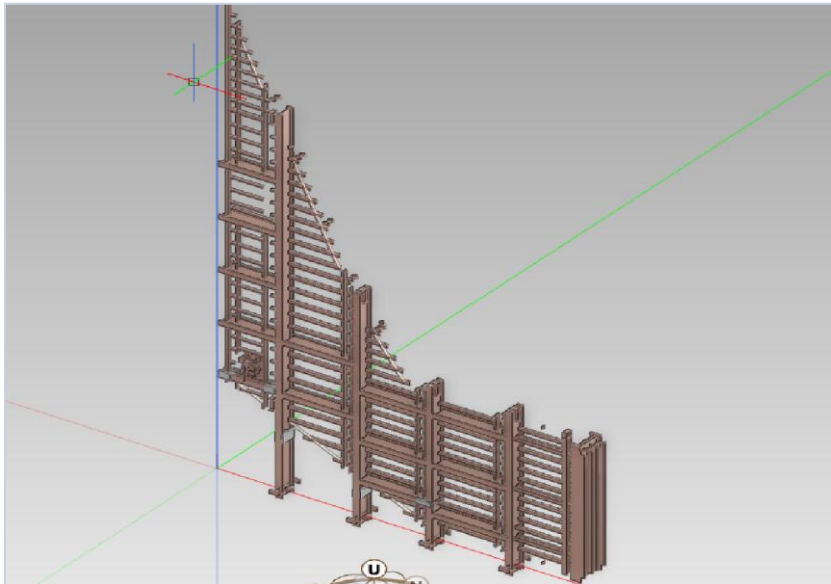
CASING OUTPUT2



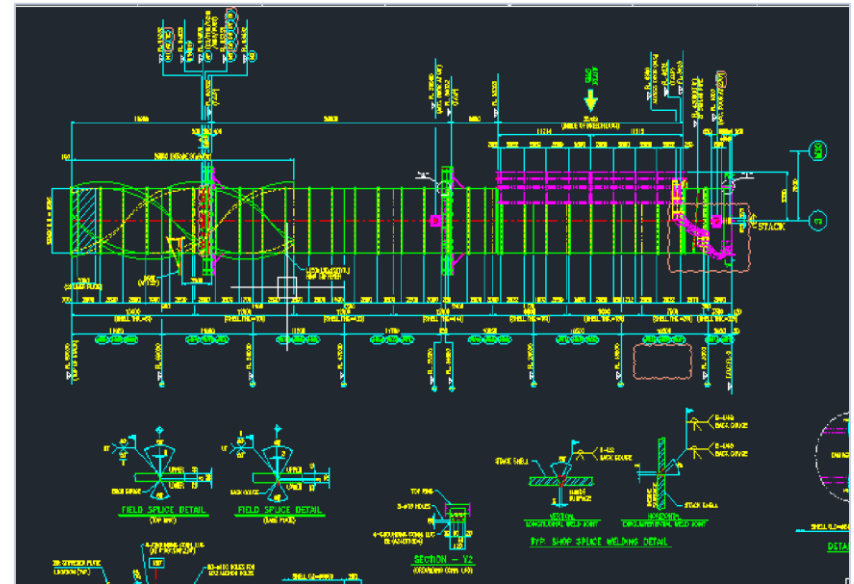
INLET DUCT INPUT1

Output Samples

Output Samples (5/7)



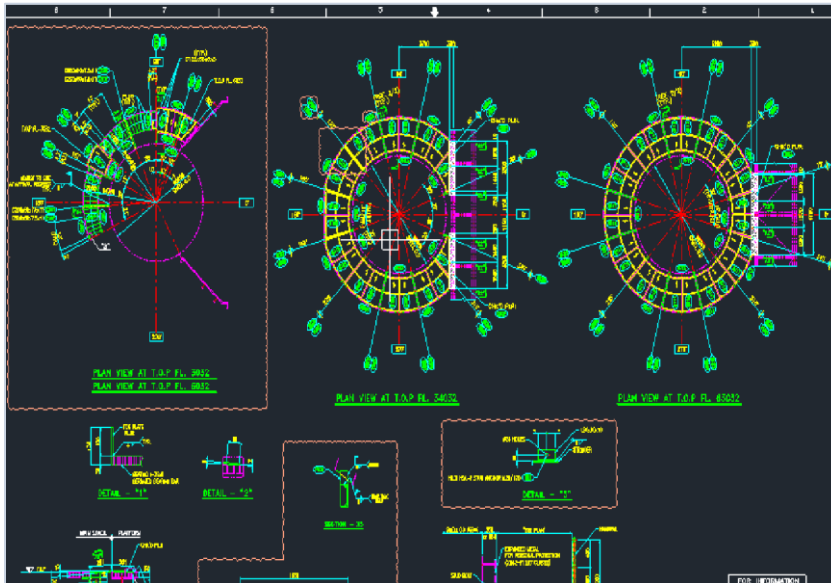
INLET DUCT OUTPUT1



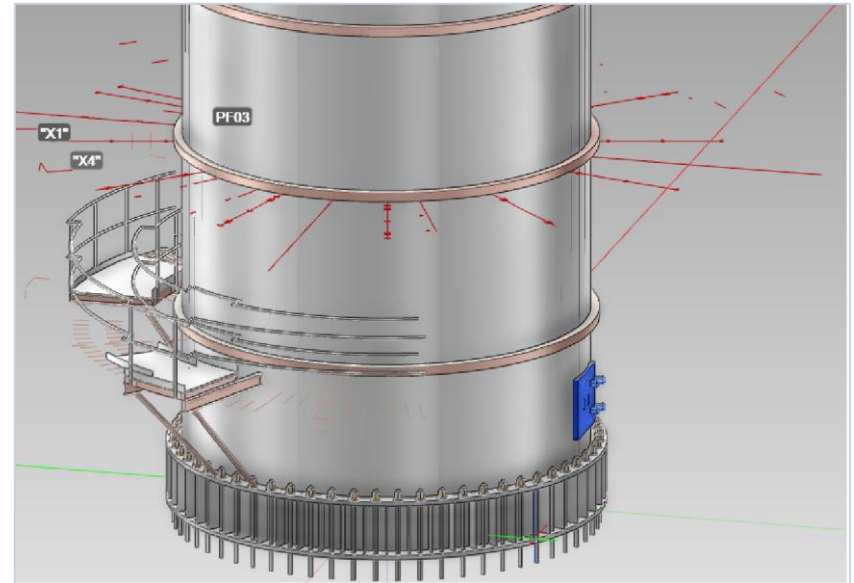
STACK INPUT1

Output Samples

Output Samples (6/7)



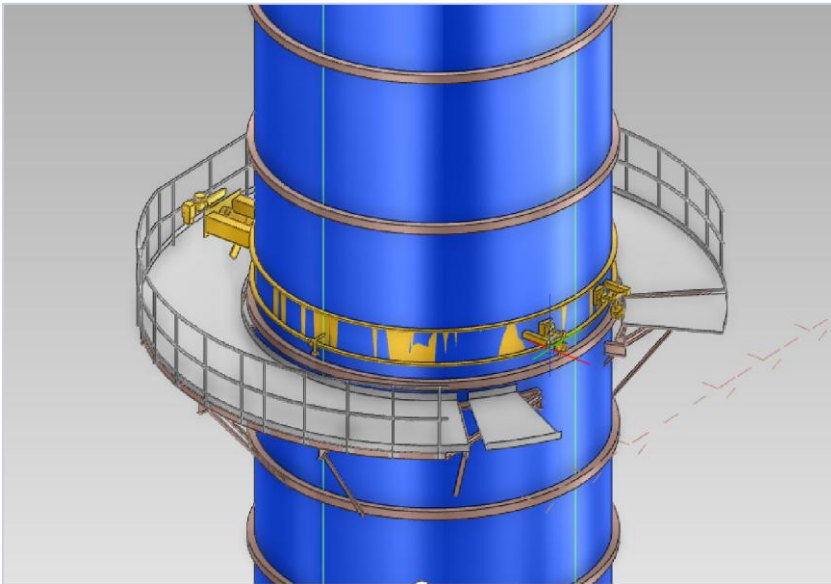
STACK INPUT2



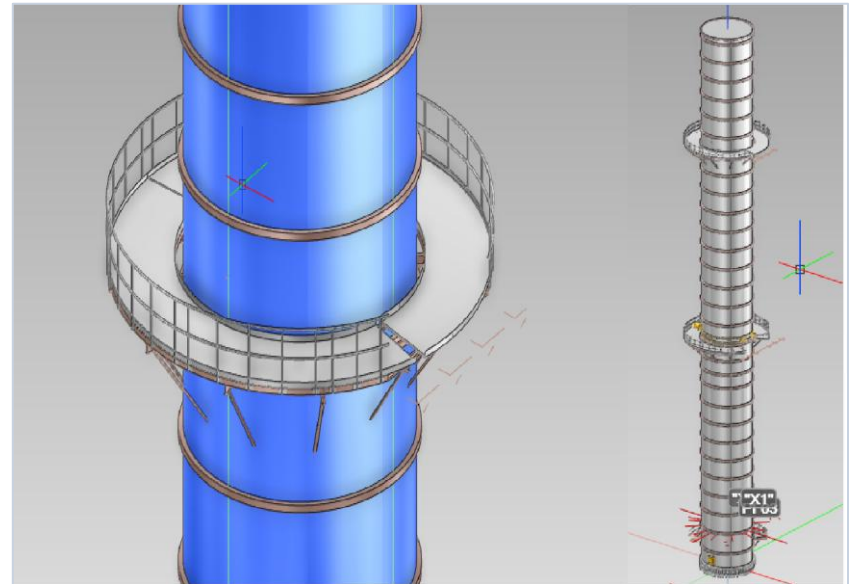
STACK OUPUT1

Output Samples

Output Samples (7/7)



STACK OUTPUT2



STACK OUTPUT3

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: HRSG 3D Auto Modeling from 2D Dwg.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: Ace_hrsg_runall.PY.
- Output samples and the program window above reflect the current build.

Thank You

HRSG 3D Auto Modeling from 2D Dwg · Plant Engineering Total Service System